Statistics is the study of data

Statistics types :→

1] Descriptive statistics: ~~types~~ ~~data~~ describes the population or dataset through numerical calculations, graphs or tables. It provide graphical summary of data
- It will tell - What does data show?
- Various methods are:
A) Central tendancy - Mean, Median, Mode
B) measures of dispersion or spread - Range, Quartile deviation, Standard deviation
c) Measure of Distribution - Skewness, Kurtosis
D) Data Presentation - Tables, Bar/Pie Charts, Histogram, freq. polygon, Scatter plots, Box plot Distribution Chart, Bubble chart etc.

2] Inferential Statistics:
- Inferential statistics makes inferences and predictions, conclusions about the data set and applied probabilities.
- It is simply used for explaining the meaning of descriptive statistics
- It is used to analyze, interpret results & draw conclusions.
- Inferential statistics offers several key methods
i) Confidence Interval
ii) Hypothesis testing - Z-test, t-test, ANOVA test
iii) Central limit theorm

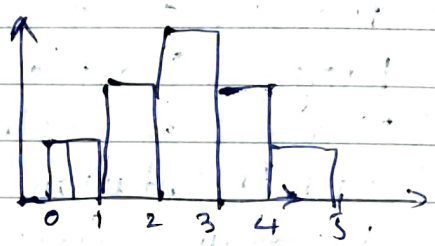
Ex: suppose e-commerce company wants to know whether a new recommendation system increases customer purchase: population: 1000000, Sample: 10,000, old system: Avg purchase: 2,500/-, New system Avg: purchase 2,800/-

A data scientist uses inferential statistics, such hypothesis test to determine whether increase from 2,500 to 2,800 is statistically significant or simply due to random ch

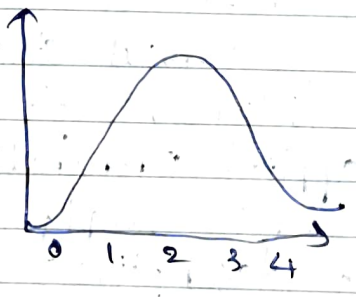
* Distribution - Probability Distribution
  - A distribution is a frequency of values in the dataset
  - A statical data distribution is a function that shows the possible values of variable and how frequently they occure. It provides a mathematical description of data's behaviour which indicate where

* Types of Distribution (Statistical data distribution)

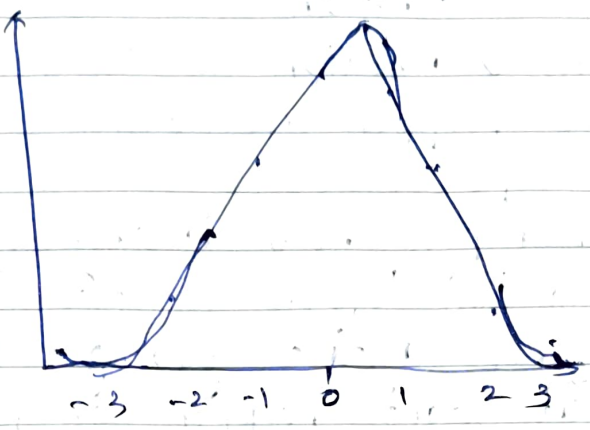
A) Discrete distribution : Where outcomes are Countable. Based on random variables which can assume only c
  - Binomial
  - poission
  - uniform
  - Bernoulli



B) Continuous Distribution : Where outcomes can take any value in a range
  - Normal
  - Exponential



1) Normal distribution
  - Normal distribution is one of the most commonly used distributions having a Symmetrical bell shaped curve.
  - In this most data points are close to mean and as you move away from it number of data points decreases.

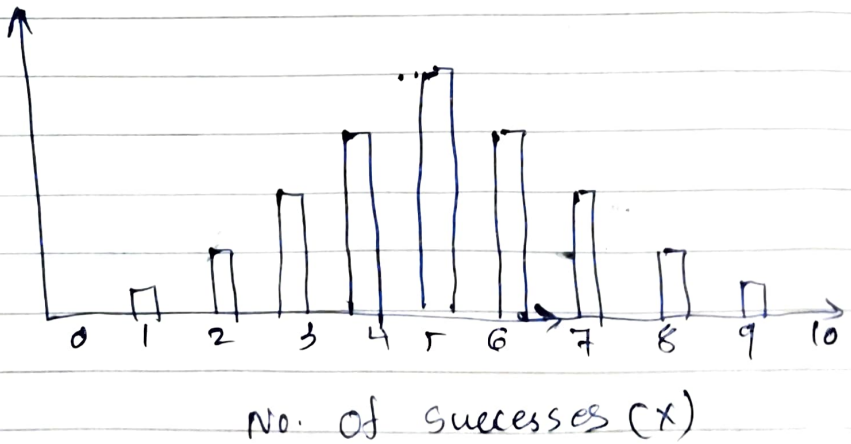


concentrated ~~& how spread~~.

~~by~~ seperated values.

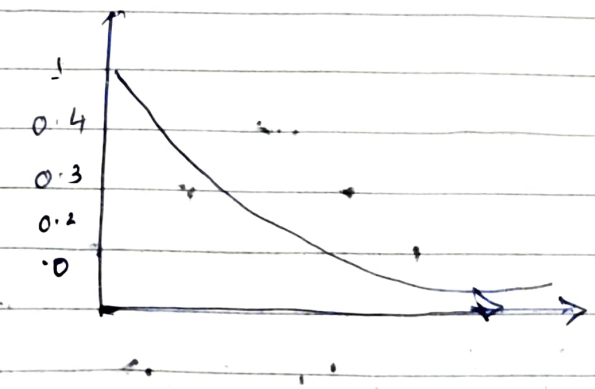
2) Binomial Distribution

- Binomial distribution is used when you perform a action multiple times and each time there are two possible outcomes sucess or failure.
- For example if you flip a coin 10 times and ~~what~~ want to know how many time you will get heads, you can apply binomial distribution to calculate that probability.
- It helps you to find the chances of getting a certain number of sucess or failures



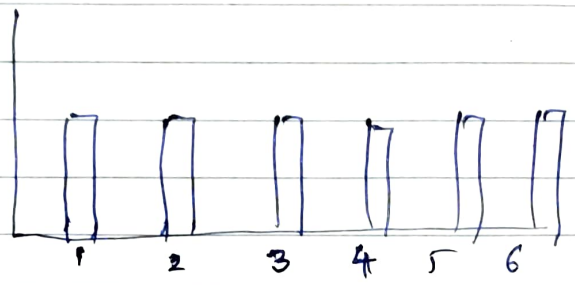
3) Poission Distribution (Diagram same as Binomial)
- Poission distribution is used to model event that happen randomly over a fixed time or space
- For example how many cars pass through a toll booth in one hour or how many emails you get in a day
- These event dont happen at regular interval
- It's useful when events are independent and occure at a constant average rate.

4) Exponential Distribution
- The exponential distribution is closely related to the poission distribution but instead of counting how many events occure within the fixed time it focuses on time between events
- for example if you are waiting for a bus, exponential distribution can help you to estimate how long you have to wait before the next bus arrives.



5) Uniform distribution

- The uniform distribution is quite straight forward.
- In this type of distribution every outcome has equal chance of occurring.
- Imagine rolling a fair six-sided die with each number 1 to 6 has an equal probability of coming up

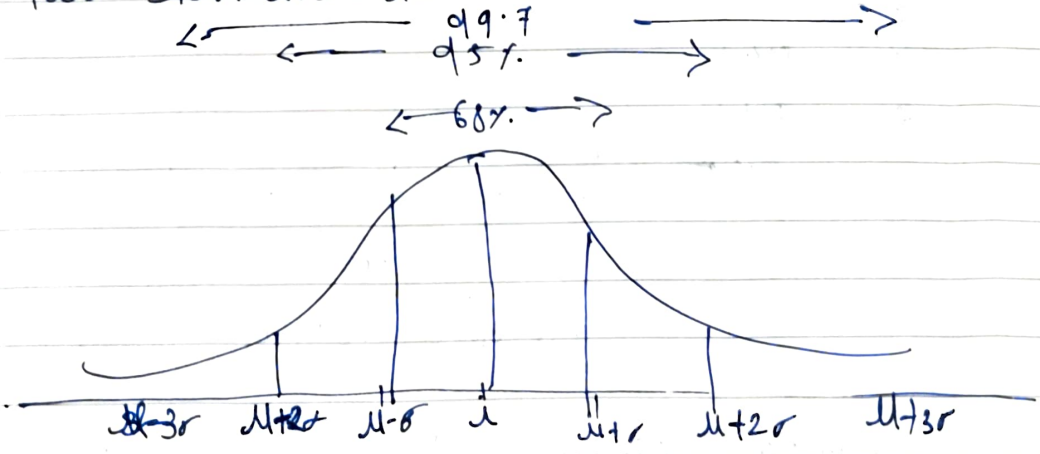


Normal Distribution

- The bell shaped normal distribution is iconic in traditional statistics
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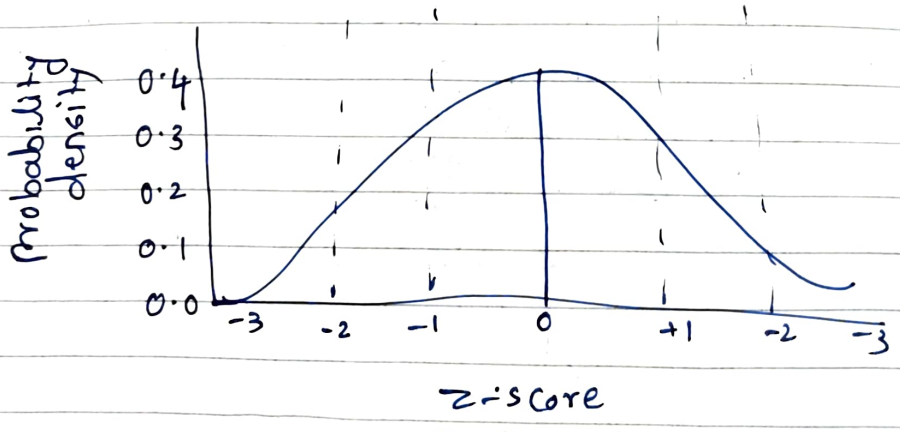
- Total area = 1 ,
- ~~Mean divides~~ Always variables used in a typical data science project may not normally distributed , but many statistics normally distributed in their sampling distribution (central limit theorm)
- The normal distribution is also reffered to as Gaussian distribution
- In normal distribution 68% of the data lies within one standard deviation
- of the mean , 95% data lies within two standard deviation



Standard normal - A normal distribution with mean = 0 & standard deviation = 1

$$\phi(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

- The standard normal distribution also called z-distribution is a special normal distribution where mean is 0 and standard deviation is 1.
- Any normal distribution can be standardized by converting to value to z score.
- z score tell you how many standard deviations from the mean to each value lie.
- Converting a normal distribution into a z-distribution allows you to calculate the probability of certain values occuring & compare different data set.



Standard normal

Z-score formula - A Z-score converts "How far is this from Avg" into "how unusual is this on a scale everyone can compare"

$$Z = \frac{X-\mu}{\sigma}$$

where X = value, μ = mean, σ - standard deviation.

Z-score → it converts raw number into relative position.

Normal Distribution

- Normal distribution is one of the most important and commonly used continuous distribution

Definition

- A continous random ~~va~~ variable X is said to follow normal distribution with parameter μ (mean) and σ^2 (std. deviation) (variance) if its probability density function is given by,

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{\frac{(x-\mu)^2}{2\sigma^2}} \quad -\infty < x < \infty$$

$-\infty < m < \infty$ & $\sigma^2 > 0$

OR

$$f(x) = \frac{1}{\sqrt{2\pi} \cdot \sigma} \cdot e^{\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

- A continuous random variable X following normal distribution with mean μ & standard deviation σ is reffered to as $X \sim N(\mu, \sigma)$

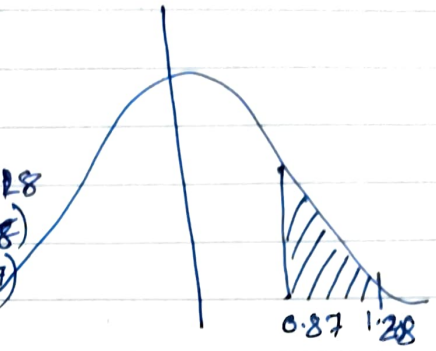
- When $\mu = 0$ & $\sigma = 1$, then $\hat{X}$ it is called Standard normal variate.

$$Z = \frac{X-\mu}{\sigma}$$

Ex·1 Find the probability that random variable having the standard normal distribution will take value between 0·87 and 1·28

Soln $P(0.87 < z < 1.28)$

= area betⁿ $z = 0.87$ to $z = 1.28$



= area from ($z = 0$ to $z = 1.28$) − area from ($z = 0$ to $z = 0.87$)

$= 0.3997 - 0.3078$

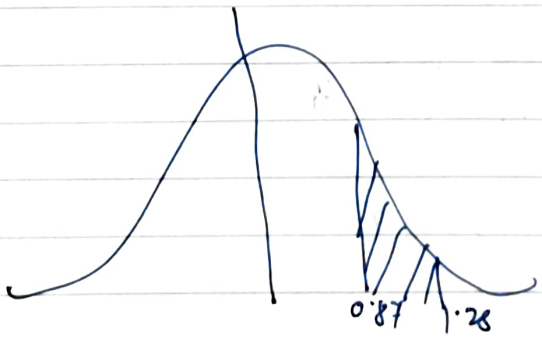
$= 0.0919$

Ex·2 Find the probability that random variable having Standard normal distribution will take value between

i) 0·87 to 1·28

ii) −0·34 and 0·62

i) 0·87 to 1·28



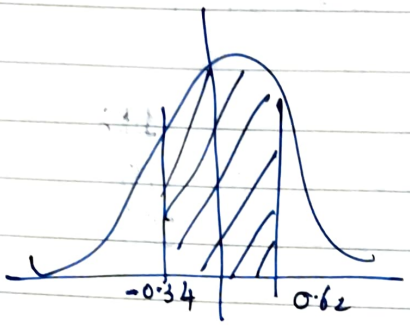
= area betⁿ $z = 0.87$ to 1.28

= are betⁿ ($z = 0$ to $z = 1.28$) − area betⁿ ($z = 0$ to $z = 0.87$)

$= 0.3997 - 0.3078 = 0.0919.$

ii) -0.34 to 0.62

= area from ($z = -0.34$ to $z = 0.62$)



= area from ($z = 0$ to $z = 0.34$) + area from ($z = 0$ to $z = 0.62$)

$= 0.1331 + 0.2324$

$= 0.3655$

Ex.3 For Normal variate with mean 2.5 and Standard deviation 3.5. Find the probability that

i) $2 \leq X \leq 4.5$

ii) $-1.5 \leq X \leq 5.3$

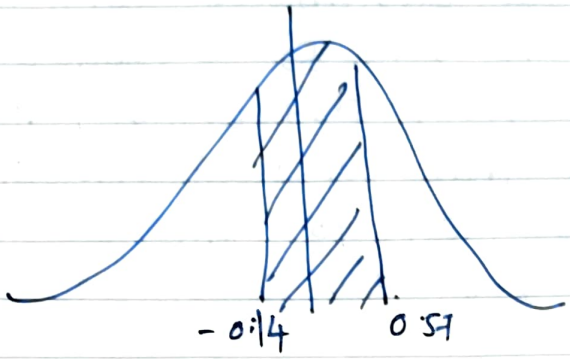
Sol^n $\mu = 2.5$, $\sigma = 3.5$

i) $P(2 \leq X \leq 4.5)$

when $X = 2$, $z = \frac{X - \mu}{\sigma} = \frac{2 - 2.5}{3.5} = 0.14$

when $X = 4.5$, $z = \frac{X - \mu}{\sigma} = \frac{4.5 - 2.5}{3.5} = 0.57$

So $p(2 \leq x < 4.5) = p(z = -0.14 \text{ to } z = 0.57)$



= Area betⁿ ($z = -0.14$ to $z = 0.57$)

= Area betⁿ ($z = -0.14$ to $z = 0$) +
Area betⁿ ($z = 0$ to $z = 0.57$)

= Area betⁿ ($z = 0$ to $z = 0.14$) +
Area betⁿ ($z = 0$ to $z = 0.57$)

$= 0.5 + 0.2157 = 0.2714$

ii) $p(x = -1.5 \leq X \leq 5.5)$

when $x = -1.5$ $\quad z = \frac{x - \mu}{\sigma} = -1.14$

when $x = 5.3$ $\quad z = \frac{x - \mu}{\sigma} = -0.8$

$p(-1.5 \leq x \leq 5.5) = p(-1.14 < z \leq 0.8)$

= Area from ($z = -1.14$ to $z = 0$) +
Area from ($z = 0$ to $z = 0.8$)

= Area from ($z = 0$ to $z = 1.14$) +
Area from ($z = 0$ to $z = 0.8$)

$= 0.3729 + 0.2881 = 0.6610$

Ex. 4 The marks obtained by students in a colle are normally distributed with mean 65 and varience 25. If 3 students are selected at random from this college, what is the probability that at least one of them would score more than 75 marks?

Soln $\mu = 65$, $\sigma^2 = 25$ so $\sigma = 5$

Step 1

We have standard normal varience

$$Z = \frac{X - m}{\sigma}$$

$\therefore$ When $X = 75$, $Z = \frac{75 - 65}{5} = 2$

Step 2 : Find P (a single student scores more than 75)

$P(X > 75) = P(Z > 2) =$

$=$ area betn $P(z > 0$ to $z > 2)$

$= 0.4772$

Since total area to right of mean $Z = 0.5$

$P(Z > 2) = 0.5 - 0.4772 = 0.0228$

So there is 0.0228 chance any one randomly chosen students score above 75.

Step 3 : Find p - a students does not score more than 75

$P(X \leq 75) = 1 - 0.0228 = 0.9772$

Step 4 : Find p - none of the 3 students score than 75

Since 3 students are independant, multiply this probabidity three times.

p (none score > 75) $= 0.9772 \times 0.9772 \times 0.9772 = 0.933$

Steps: Find p : at least one scores more than 75

At least one is complement of one

p (at least one > 75) $= 1 - 0.933 = 0.067$

Piossion Distribution

- Piossion distribution is the limiting case of binomial distribution under the following condition

i) The trials (n) is infinitely large
   i.e. $n \to \infty$

ii) p, the probability of success in each trial is constant and infinitely small
   i.e $p \to 0$

iii) np, the average success is finite say m, i.e $np = m$

iv) A trial results in only two ways - success or failure

v) If p & q are probabilities of success and failure, then $p + q = 1$

vi) mean = variance $\Rightarrow \mu = \sigma^2$

Poision distribution application

i) the number of deaths due to disease such as heart attack, cancer etc

ii) The number of accidents during a week or month etc.

iii) the number of phone calls received at a particular telephone exchange during period of time.

iv) The number of cars passing a particular point on a road during a period of time

v) The number of printing mistakes on a page of a book.

$e = 2.718$

The poisson distribution is

- A random variable x is said to follow poisson distribution if the probability of x is given by

$$P(X=x) = \frac{e^{-\mu}\mu^{x}}{x!}, \quad x = 0, 1, 2 \cdots$$

Ex 1 ~~find out fallacy~~ If the mean of the poisson distribution is 4. Find $P(m-2\sigma < X < m+2\sigma)$

Solut^{n} For poisson distribution mean = variance

$$\mu = \sigma^2$$

$$\mu = 4 \quad \text{so} \quad \sigma = 2$$

$$P(m-2\sigma < X < m+2\sigma) = P(0 < X < 8)$$

$$= P = [X = 1, 2, 3 \cdots 7]$$

$$P(X) = \frac{e^{-\mu}\mu^{x}}{x!}$$

$$= e^{-4}\frac{4^{x}}{x!}$$

$$= e^{-4}\left[\frac{4}{1!} + \frac{4^2}{2!} + \frac{4^3}{3!} + \frac{4^4}{4!} + \frac{4^5}{5!} + \frac{4^6}{6!} + \frac{4^7}{7!}\right]$$

$$= 0.93$$

Ex. 2 A hospital switch board receives an average of 4 emergency calls in a 10 minutes interval. What is probability that
i) there are 2 emergency calls
ii) there are exactly 3 emergency calls in a interval of 10 minutes.

Sol^n

We have $p(x) = \frac{e^{-\mu} \cdot \mu^x}{x!}$

Here $\mu = 4$

i) $P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$

$$= \frac{e^{-4} 4^0}{0!} + \frac{e^{-4} 4^1}{1!} + \frac{e^{-4} 4^2}{2!}$$

$$= e^{-4} (1 + 4 + 8) = 0.238$$

ii) $P(X=3) = \frac{e^{-4} 4^3}{3!} = 0.195$